Book 1 — Lecture 2 — Exercise 5

Derivatives of sin, cos, exp, and ln

Prompt. Prove each of the formulas:

$$\frac{d}{dt}\sin(t) = \cos(t), \qquad \frac{d}{dt}\cos(t) = -\sin(t), \qquad \frac{d}{dt}e^t = e^t, \qquad \frac{d}{dt}\ln(t) = \frac{1}{t}.$$

Hint: Look up trigonometric identities and limit properties in a reference book.[1]

Two standard limits

The sine and cosine proofs use

$$\lim_{h\to 0}\frac{\sin h}{h} = 1 \qquad \text{and} \qquad \lim_{h\to 0}\frac{\cos h - 1}{h} = 0.$$

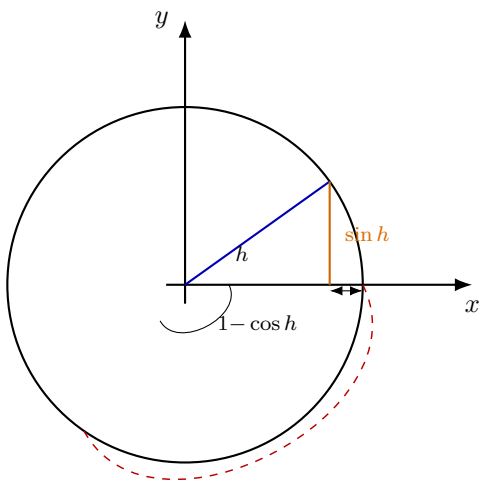
The second follows from the first via the identity $1 - \cos h = 2\sin^2(h/2)$:

$$\frac{1-\cos h}{h} = \frac{2\sin^2(h/2)}{h} = \underbrace{\left(\frac{\sin(h/2)}{h/2}\right)^2}_{\to 1}\cdot\frac{h}{2} \to 0,$$

hence $\dfrac{\cos h - 1}{h} \to 0$.

Picture (small-angle limits). On the unit circle, a small central angle h (radians) cuts off an arc of length h and a vertical segment of length $\sin h$. For $h \to 0^+$, arc and segment nearly coincide, so $\sin h/h \to 1$. The point on the circle lies at $(\cos h, \sin h)$, so the horizontal gap $1 - \cos h$ is much smaller than h; that is the geometric content of $(\cos h - 1)/h \to 0$.

[1] All limits below are one-sided at domain boundaries where needed ($t > 0$ for $\ln t$). Angles are in radians.



Angle exaggerated for readability; the limits use $h \to 0$.

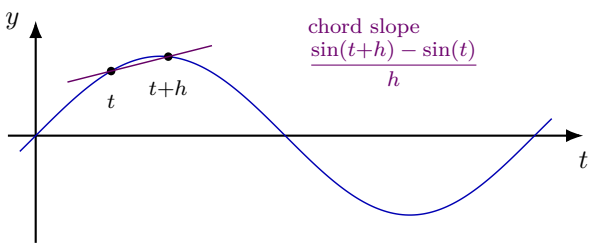
$\dfrac{d}{dt}\sin(t) = \cos(t)$

For $h \neq 0$, the addition formula $\sin(t+h) = \sin t \cos h + \cos t \sin h$ gives

$$\frac{\sin(t+h) - \sin(t)}{h} = \sin t \, \frac{\cos h - 1}{h} + \cos t \, \frac{\sin h}{h}.$$

Let $h \to 0$. The first term $\to 0$ and the second $\to \cos t$, so $\dfrac{d}{dt}\sin(t) = \cos(t)$.

Picture (sine). The Newton quotient $(\sin(t{+}h) - \sin(t))/h$ is the slope of the chord joining two points on $y = \sin t$. As $h \to 0$, that chord approaches the tangent line, whose slope is $\cos t$.



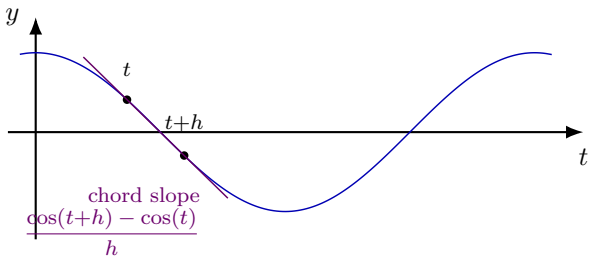
The illustration fixes a sample t, h; the proof is the limit $h \to 0$.

$\dfrac{d}{dt}\cos(t) = -\sin(t)$

Using $\cos(t+h) = \cos t \cos h - \sin t \sin h$,

$$\frac{\cos(t+h) - \cos(t)}{h} = \cos t \, \frac{\cos h - 1}{h} - \sin t \, \frac{\sin h}{h} \longrightarrow -\sin t.$$

Picture (cosine). The same chord idea applies to $y = \cos t$. Here the chord slope tends to $-\sin t$ (downhill where $\sin t > 0$).



Again, a fixed sample (t, h) for drawing; the derivative is the limit.

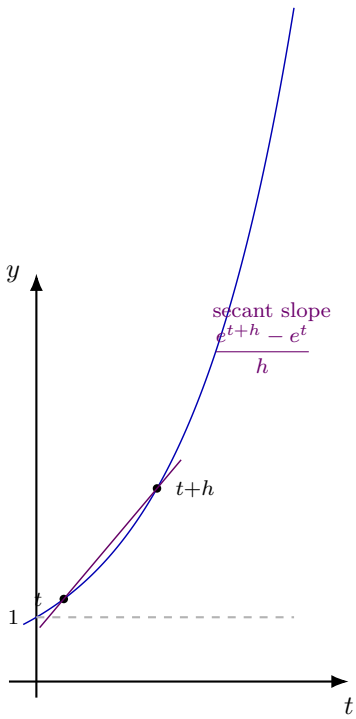
$\dfrac{d}{dt}e^t = e^t$

For every $h \neq 0$,

$$\frac{e^{t+h} - e^t}{h} = e^t \cdot \frac{e^h - 1}{h}.$$

It suffices to show $\lim_{h\to 0} \dfrac{e^h - 1}{h} = 1$.[2] Then $\dfrac{d}{dt}e^t = e^t \cdot 1 = e^t$.

Picture (exponential). The graph $y = e^t$ is self-similar in vertical scale: shifting t by h multiplies heights by e^h. Thus $(e^{t+h} - e^t)/h = e^t \cdot (e^h - 1)/h$: the secant slope at height e^t is e^t times the normalized slope $(e^h - 1)/h$ through $(0, 1)$.



Vertical stretch factor e^t links this secant to $(e^h - 1)/h$ at the reference point $(0, 1)$.

[2] Many texts define e^x first (series, integral, or limit) and derive this limit; the lecture hint points to a reference for whichever route your edition uses. One common route: with $e^x = \sum_{n=0}^{\infty} x^n/n!$, one has $e^h - 1 = h + h^2/2! + \cdots = h(1 + O(h))$, so $(e^h - 1)/h \to 1$.

$\dfrac{d}{dt}\ln(t) = \dfrac{1}{t}$ for $t > 0$

Fix $t > 0$ and take $h \neq 0$ small enough that $t + h > 0$. Then

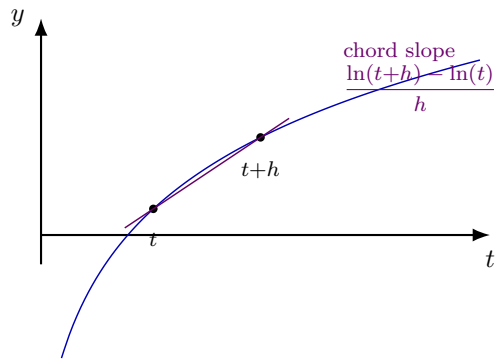
$$\frac{\ln(t+h) - \ln(t)}{h} = \frac{1}{h}\ln\left(1 + \frac{h}{t}\right) = \frac{1}{t} \cdot \frac{\ln(1+u)}{u}, \qquad u = \frac{h}{t}.$$

As $h \to 0$, also $u \to 0$. If $v = \ln(1+u)$, then $u = e^v - 1$ and $v \to 0$ as $u \to 0$, and

$$\frac{\ln(1+u)}{u} = \frac{v}{e^v - 1} = \left(\frac{e^v - 1}{v}\right)^{-1} \longrightarrow 1$$

using the same exponential limit $\dfrac{e^v - 1}{v} \to 1$ as above. Therefore the Newton quotient tends to $\dfrac{1}{t}$.

Picture (logarithm). On $y = \ln t$, a step $\Delta t = h$ produces a rise $\ln(t+h) - \ln(t)$. Writing $u = h/t$, the relative step along the horizontal axis is u; the quotient $\ln(1+u)/u$ is the slope of the chord from $(1, 0)$ to $(1+u, \ln(1+u))$ in a *normalized* picture, and it tends to 1 as $u \to 0$. Rescaling by $1/t$ gives slope $1/t$ at t.



Domain $t > 0$; the algebra rewrites this slope as $\dfrac{1}{t} \cdot \dfrac{\ln(1+u)}{u}$ with $u = h/t$.

Result. For all real t (in radians),

$$\frac{d}{dt}\sin(t) = \cos(t), \qquad \frac{d}{dt}\cos(t) = -\sin(t).$$

For all real t, $\dfrac{d}{dt}e^t = e^t$. For $t > 0$, $\dfrac{d}{dt}\ln(t) = \dfrac{1}{t}$.

Example. Check at $t = 0$: $\cos'(0) = -\sin(0) = 0$; $\dfrac{d}{dt}\ln(t)$ is not defined at $t = 0$ because ln is undefined there.